



Analog IC Design

CMOS Technology

MOHAMMED NASR EL-DIN

Section 3 - Lab 06

Differential amplifier design

Sizing

Note that $I_{SS}R_D = 2V_{ocm}$ must be smaller than $(V_{DD} - V_{dsat-tail})$ for proper large signal characteristics (why?).

To keep the tail current source in saturation in case **current steering**.

Choose R_D to meet the CM output level spec.

$$I_{SS}R_D = 2V_{ocm} = 2 \times 0.6v = 1.2v \Rightarrow R_D = \frac{1.2v}{I_{SS}} = \frac{1.2v}{40\mu A} = 30K\Omega$$

Input pair sizing

$$I_D = 20\mu A$$

$$\frac{g_m}{I_D} = \frac{A_{vd}}{0.91 \times V_{RD}} = \frac{8}{0.91 \times 0.6v}$$

$$V_{DS} \text{ of the tail CS} = 0.3 \Rightarrow V_S \text{ of the input transistor} = 1.8 - 0.3 = 1.5v$$

$$\Rightarrow |V_{SB}| = 1.8 - 1.5 = 0.3v \text{ and } |V_{DS}| = V_S - V_{RD} = 1.5 - 0.6 = 0.9v$$

$$\Rightarrow L = 350nm, W = 30\mu m$$

ID	20u
gm/ID	8/(0.91*0.6)
ro	10*0.6/ID
VDS	0.9
VSB	0.3
Stack	1

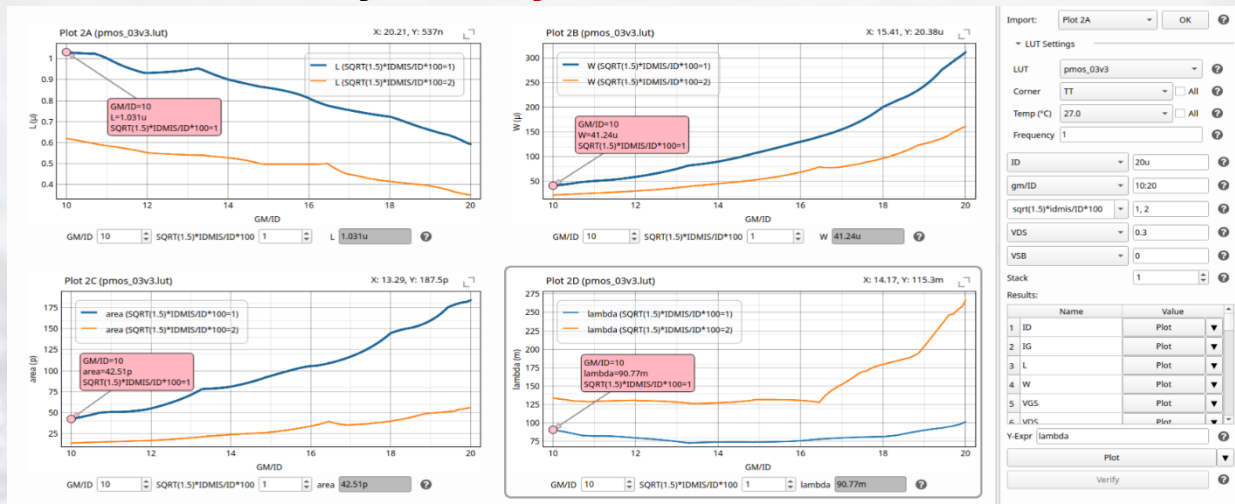
Name	TT-27.0
1 ID	20u
2 IG	N/A
3 L	350n
4 W	30u

Given the above assumption for V_{DS} of the tail current source, calculate the required CM input level.

$$\text{From ADT: } V_{GS} = 0.94v \Rightarrow CMIL = V_{DD} - V_{DS} - V_{GS} = 1.8 - 0.3 - 0.94 = 0.56v$$

Current mirror sizing

For compliance voltage $\leq 200mv \rightarrow \frac{g_m}{I_D} \geq 10$, take $\frac{g_m}{I_D} = 10$ to minimize the area. $V_{SB} = 0, V_{DS} = 0.3v, I_D = 20\mu A$



Simple current mirror sizing

In order to satisfy both mismatch and lambda requirements, which design curve will you pick, the 1% or the 2% mismatch?

We will pick **1%** mismatch, since as shown in the λ graph above for 2% mismatch it doesn't have a value for λ that meet the specs that is $\lambda < 0.1$

$$\Rightarrow L = 1\mu m, W = 40\mu m, \text{ for mirror transistor } W = 80\mu \text{ to get mirroring ratio} = 2$$

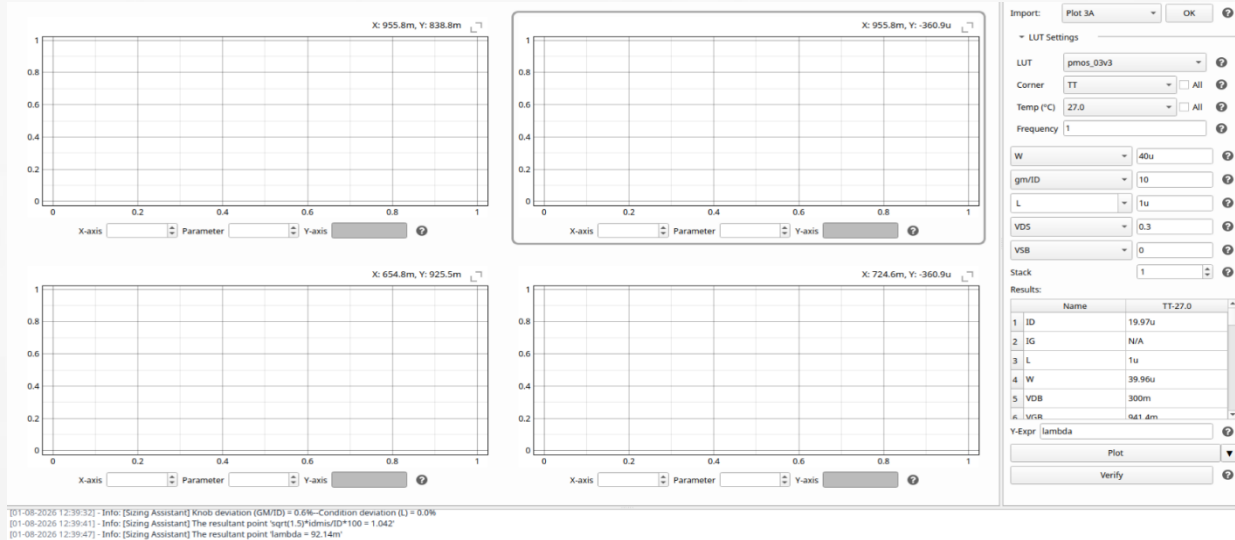
Calculate the min and max CM input levels. Is the previously selected CM input level in the valid range?

From ADT: $|V_{dsat-tail}| = 0.15v$, $|V_{GS-in}| = 0.94v$, $|V_{th-in}| = 0.87v$

$V_{DD} - |V_{dsat}| - V_{GS} = 1.8 - 0.15 - 0.94 \geq V_{icm} \Rightarrow V_{icm-max} = 0.71$

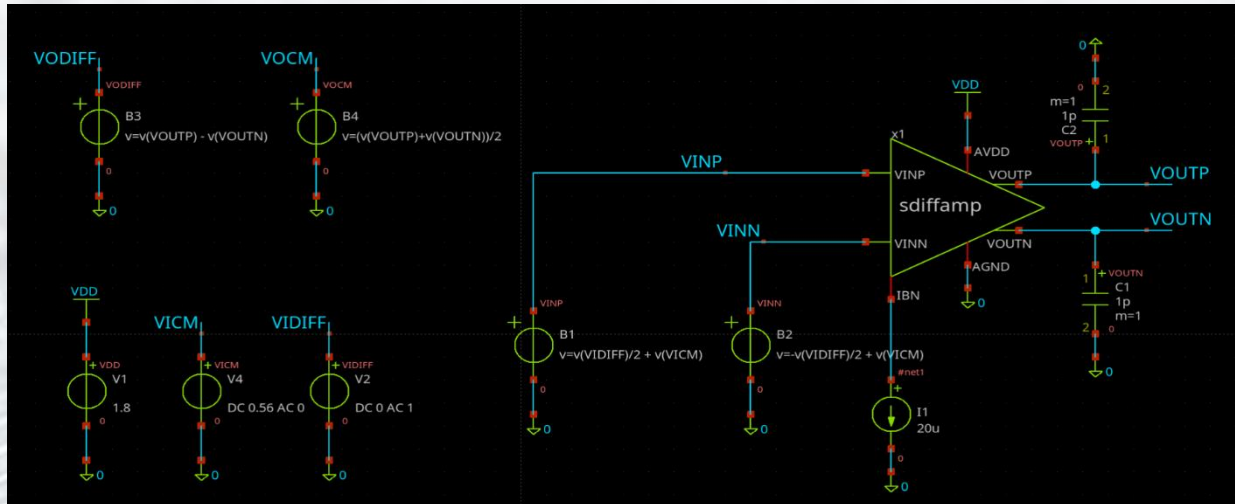
$V_{icm} \geq -|V_{th}| + V_{R_D} = -0.87 + 0.6 \Rightarrow V_{icm-min} = -0.27v$

The selected $CMIL = 0.56v$ is in the $CMIR (-0.27 \rightarrow 0.71v)$



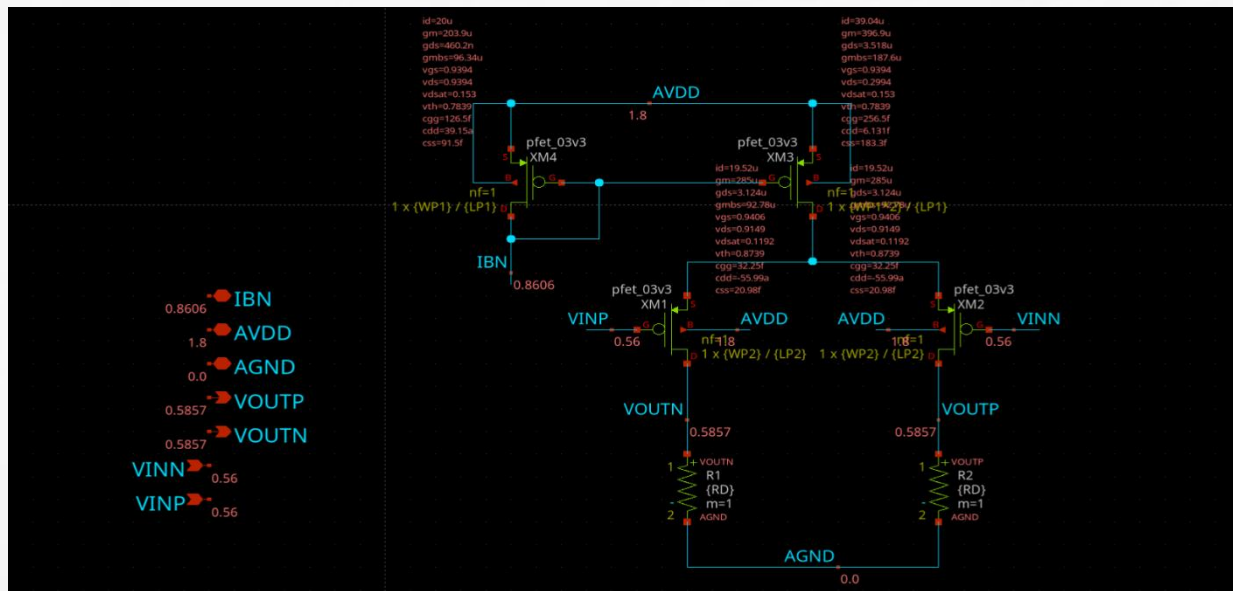
Simple current mirror design point

At the selected sizing the mismatch is almost 1% and $\lambda = 0.092 < 0.1$ which is meet our specs.



OP analysis

OP simulation



=== MOSFET Operating Points ===			
BSIM4v5: Berkeley Short Channel IGFET Model-4			
device	m.x1.xm4.m0	m.x1.xm3.m0	m.x1.xm2.m0
model	pfet_03v3.13	pfet_03v3.12	pfet_03v3.12
id	2e-05	3.90442e-05	1.95221e-05
vgs	0.939381	0.939381	0.940606
vds	0.93938	0.299392	0.914942
vth	0.783939	0.783941	0.873907
vdsat	0.152971	0.15297	0.119187
gm	0.000203874	0.000396868	0.000285029
gds	4.60249e-07	3.51789e-06	3.12392e-06
gmbs	9.63444e-05	0.000187567	9.27817e-05
BSIM4v5: Berkeley Short Channel IGFET Model-4			
device	m.x1.xm1.m0		
model	pfet_03v3.12		
id	1.95221e-05		
vgs	0.940606		
vds	0.914942		
vth	0.873907		
vdsat	0.119187		
gm	0.000285029		
gds	3.12392e-06		
gmbs	9.27817e-05		

$$M1, M2: V_{GD} = V_{GS} - V_{DS} = 0.94 - 0.91 = 0.03 < (V_{th} = 0.87)$$

$$M3: V_{GD} = V_{GS} - V_{DS} = 0.94 - 0.3 = 0.64 < (V_{th} = 0.78)$$

$$M4: V_{GD} = V_{GS} - V_{DS} = 0.94 - 0.94 = 0 < (V_{th} = 0.78)$$

⇒ All transistors operate in **saturation region**.

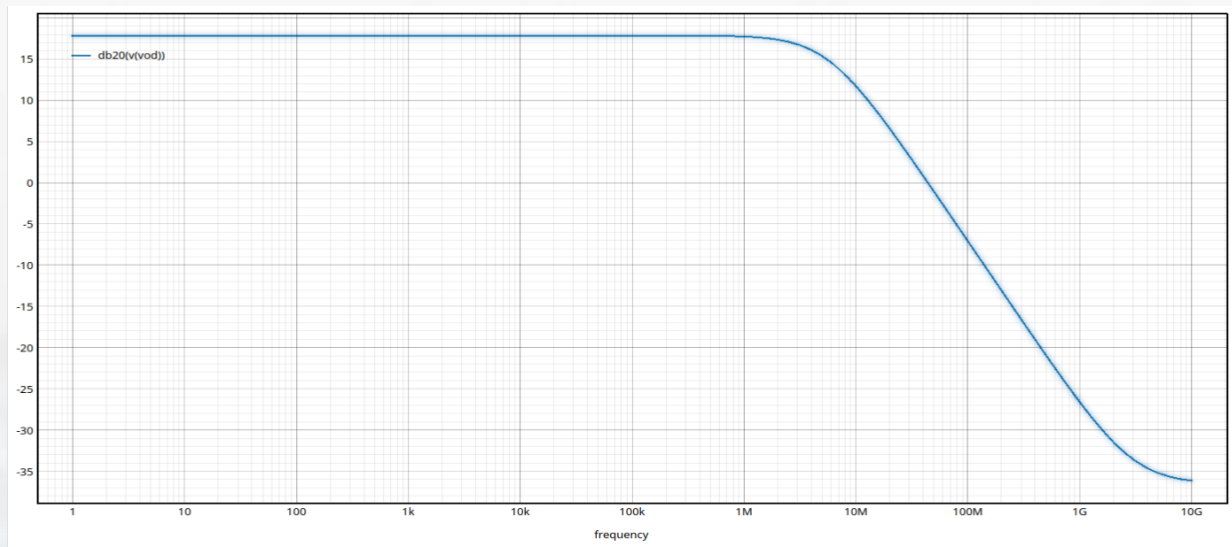
=== Bias Points ===	
M1:	gm/id = 14.6003 S/A, Vstar = 0.136983 V
M2:	gm/id = 14.6003 S/A, Vstar = 0.136983 V
M3:	gm/id = 10.1646 S/A, Vstar = 0.196762 V
M4:	gm/id = 10.1937 S/A, Vstar = 0.1962 V

AC analysis

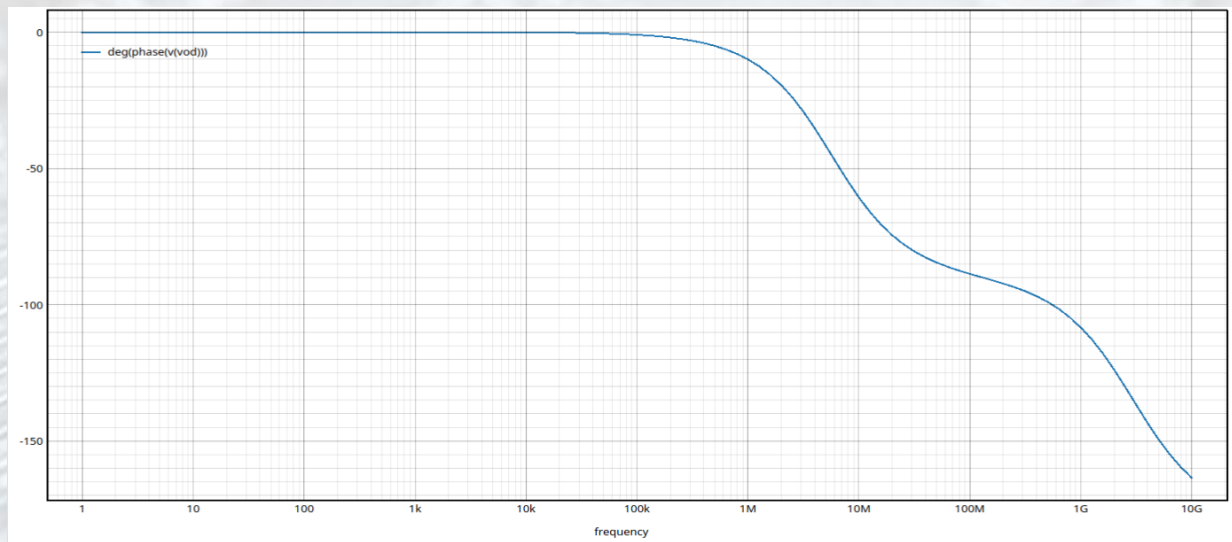
```
av_diff_db      = 1.786197e+01
av_diff_mag     = 7.818047e+00
bw_diff         = 5.683484e+06
av_cm_db        = -2.811558e+01
av_cm_mag       = 3.928447e-02
bw_cm           = 6.969060e+08

=== AC Results ===
Av Diff: 17.862 dB, BW: 5.68348E+06 Hz
Av CM: -28.1156 dB, BW: 6.96906E+08 Hz
CMRR: 45.9775 dB
```

Diff small signal



Bode plot: A_{vd} mag



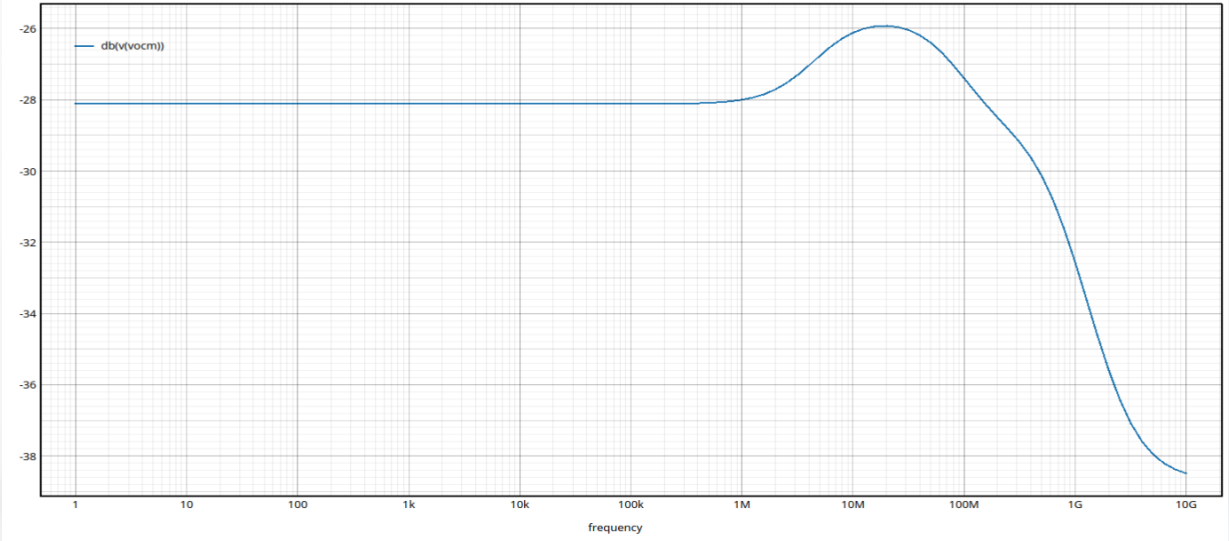
Bode plot: A_{vd} phase

Compare the DC diff gain and BW with hand analysis in a table.

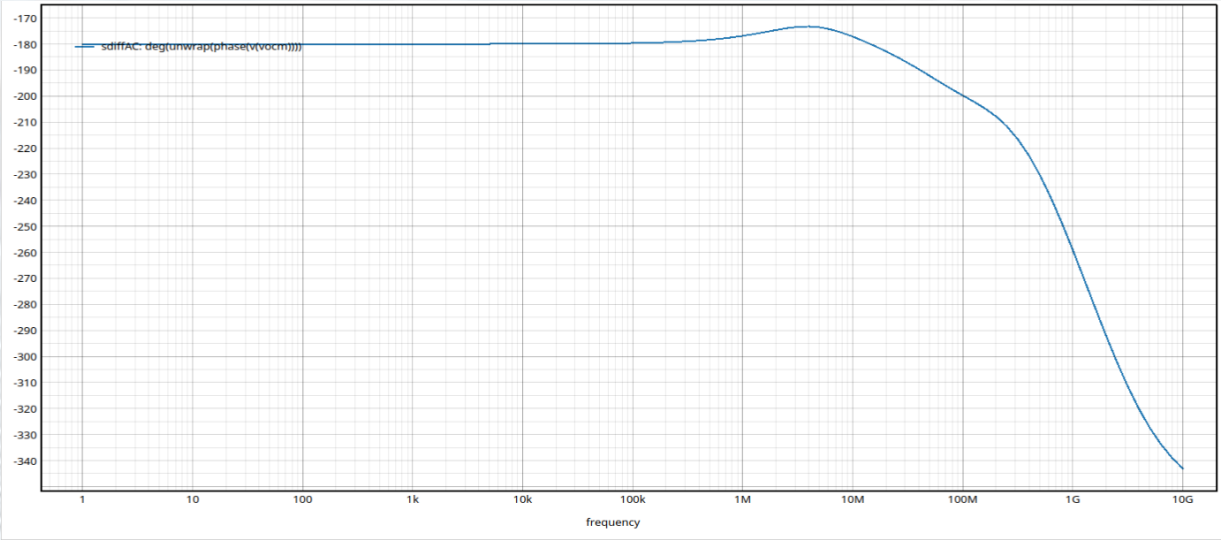
$$R_{out} = R_D // r_o (1 + g_m R_S) \approx R_D = 30K\Omega, C_{out} \approx C_L \Rightarrow BW \approx \frac{1}{2\pi R_D C_L} = \frac{1}{2\pi 30K\Omega 1pF} \approx 5.3MHz$$

parameter	Differential small signal analysis		
	Simulation	analytical	Error
gain A_{vd}	7.81	8	0.19
BW	5.7MHz	5.3MHz	0.4MHz

CM small signal



Bode plot: A_{vcm} mag



Bode plot: A_{vcm} phase

Compare the DC CM gain with hand analysis in a table. Is it smaller than “1”? Why?

$$|A_{vcm}| = \frac{g_{m1}R_D}{1+2(g_{m1}+g_{mb1})R_{ss}}, R_{ss} = r_{o3} = \frac{1}{g_{ds3}} \approx \frac{1}{3.5\mu S} \Rightarrow |A_{vcm}| = \frac{285\mu S \cdot 30K\Omega}{1+\frac{2(285\mu S+90.28\mu S)}{3.5\mu S}} \approx 0.03968$$

parameter	CM small signal analysis		
	Simulation	analytical	Error
gain $ A_{vcm} $	0.03928	0.03968	0.0004

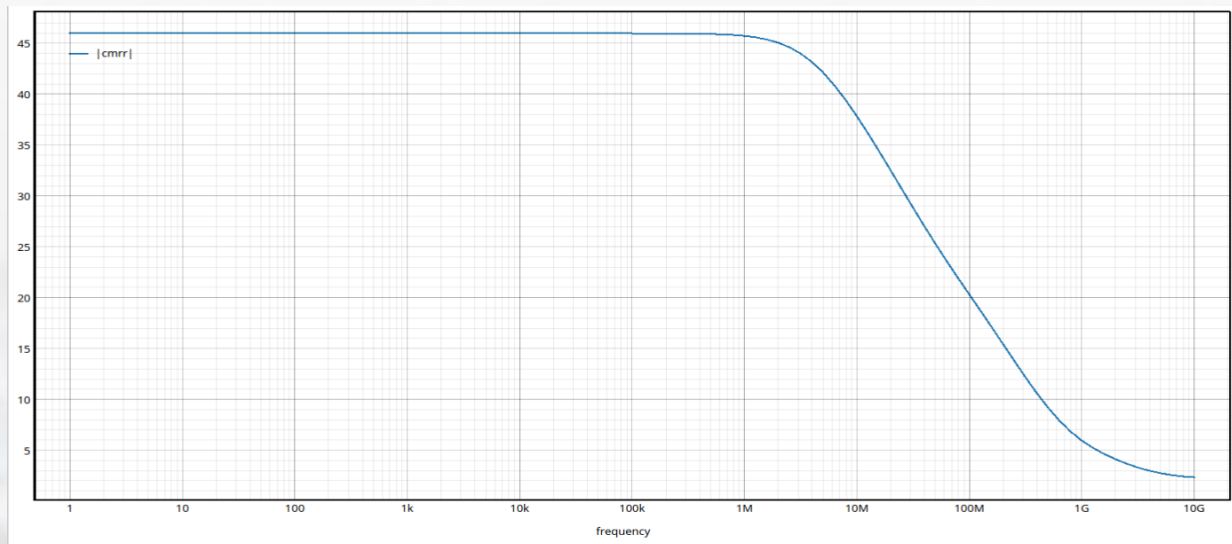
The common mode gain $|A_{vcm}| = \frac{g_{m1}R_D}{1+2(g_{m1}+g_{mb1})R_{ss}}$ is smaller than 1 due to large **degeneration** resistance R_{ss} .

Justify the variation of A_{vcm} vs frequency.

At low frequency, the common mode gain is very low due to large degeneration resistance.

At high frequency, the parasitic capacitance shunts R_{ss} decreasing the degeneration impedance causing gain rise.

At higher frequency, the output pole reduce the gain.



CMRR

Justify the variation of $\frac{A_{vd}}{A_{vcm}}$ with frequency.

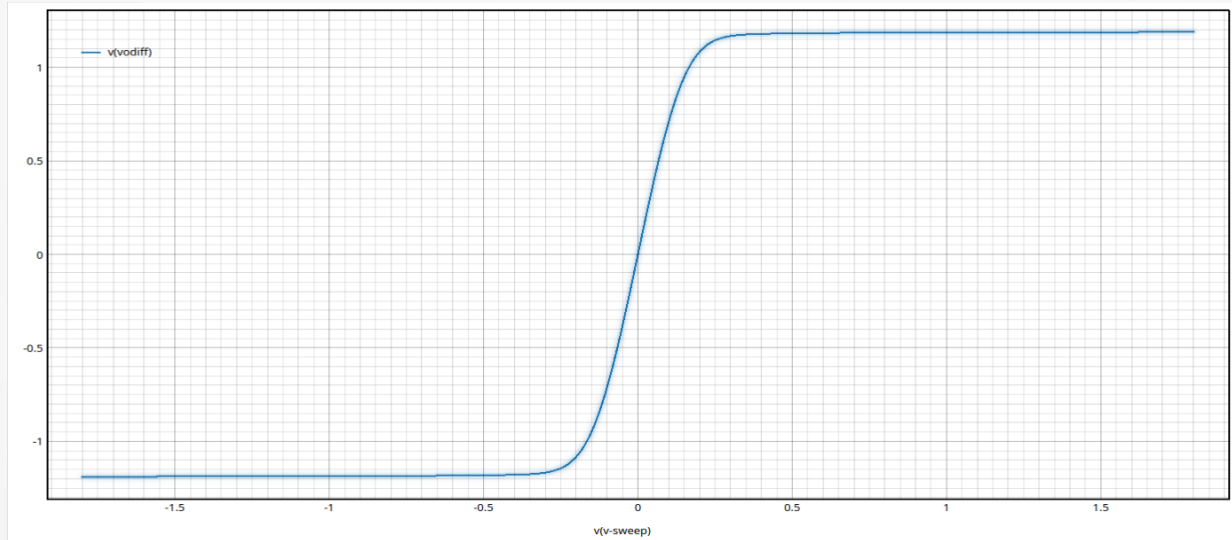
$$CMRR = \frac{A_{vd}}{A_{vcm}} = \frac{-g_{m1}R_D}{\left(\frac{-g_{m1}R_D}{1+2(g_{m1}+g_{mb1})R_{ss}}\right)} \approx 2(g_{m1}+g_{mb1})R_{ss}$$

At low frequency, the common mode rejection ratio is very high due to large degeneration resistance.

At high frequency, the common mode rejection ratio reduced due to shunt parasitic capacitance that reduce the degeneration impedance.

DC analysis

Diff large signal



V_{od} vs V_{id}

Diff Large Signal Analysis Results:
Max Output: 1.18824 V
Min Output: -1.18824 V

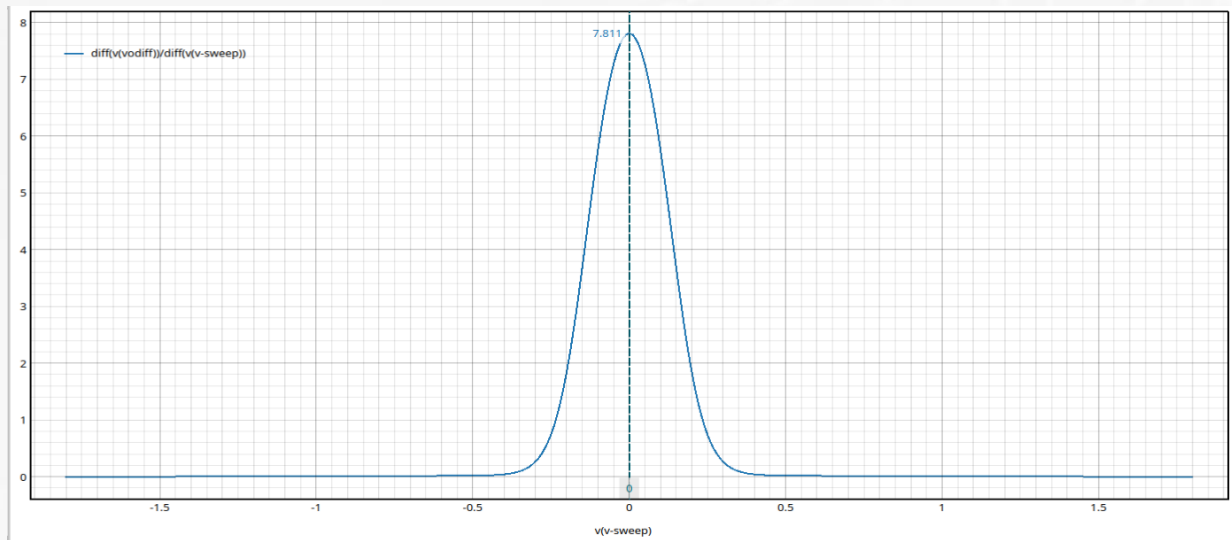
Compare the extreme values with hand analysis in a table.

At $V_{id} = V_{inP} - V_{inN} = 0 \Rightarrow V_{inP} = V_{inN} \Rightarrow I_{D1} = I_{D2} = 20\mu A \Rightarrow V_{outP} = V_{outN} \Rightarrow V_{od} = V_{outP} - V_{outN} = 0$

At **high** $V_{id} \Rightarrow V_{inP} \uparrow\uparrow, V_{inN} \downarrow\downarrow \Rightarrow I_{D1} = 40\mu A, I_{D2} = 0 \Rightarrow V_{outP} = 40\mu A \cdot 30K\Omega = 1.2v, V_{outN} = 0$
 $\Rightarrow V_{od} = V_{outP} - V_{outN} = 1.2v$

At **low** $V_{id} \Rightarrow V_{inP} \downarrow\downarrow, V_{inN} \uparrow\uparrow \Rightarrow I_{D1} = 0, I_{D2} = 40\mu A \Rightarrow V_{outP} = 0, V_{outN} = 40\mu A \cdot 30K\Omega = 1.2v$
 $\Rightarrow V_{od} = V_{outP} - V_{outN} = -1.2v$

parameter	Differential large signal analysis		
	Simulation	analytical	Error
Max output	1.19v	1.2v	0.01
Min output	-1.19v	-1.2v	0.01



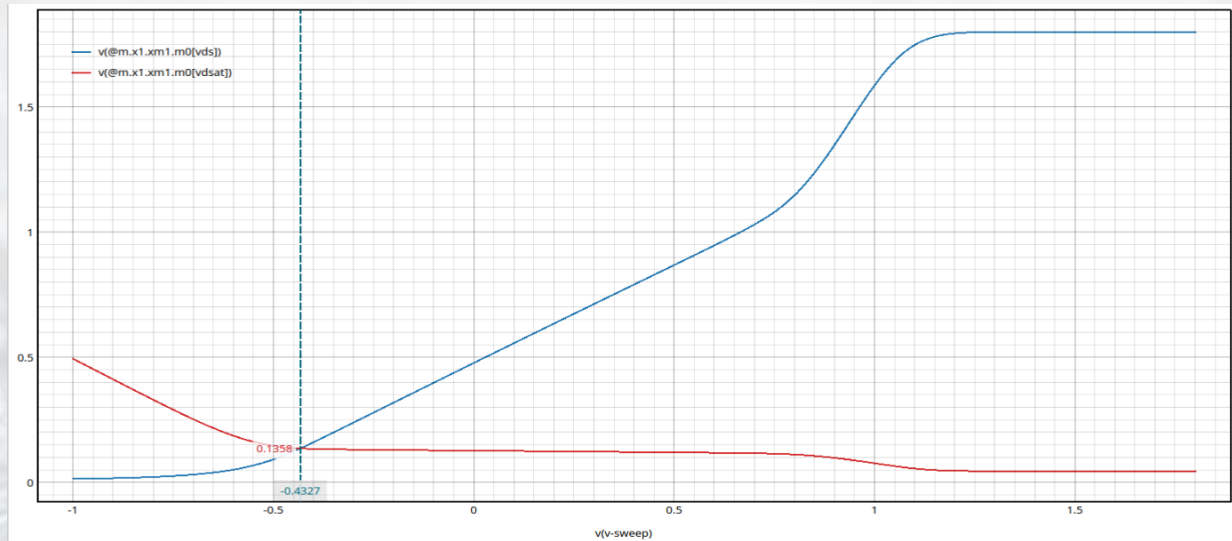
V_{od} derivative vs V_{id}

What is the peak value? Why?

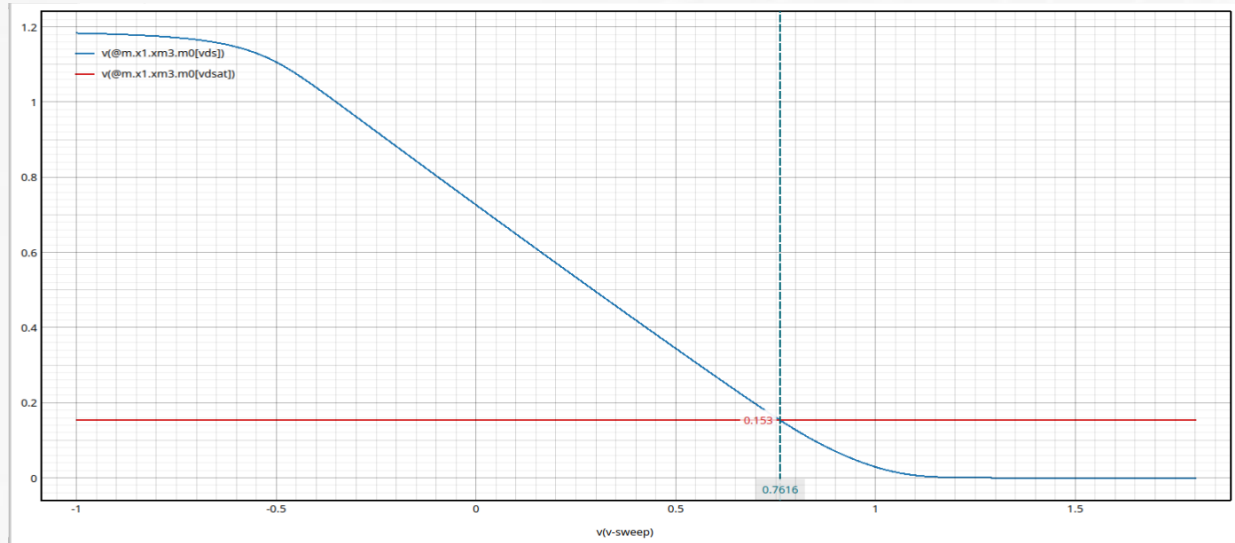
the peak value 7.81 at $V_{id} = 0$

because the derivative of V_{od} with V_{id} is the differential gain, the peak value equal max gain at $V_{id} = 0$

CM large signal: region vs V_{icm}



V_{DS} and V_{dsat} vs V_{icm} for the input pair



■ V_{DS} and ■ V_{dsat} vs V_{icm} for the tail current source

Find the CM input range (CMIR). Compare with hand analysis in a table.

When $V_{icm} \uparrow$, constant $I_D \Rightarrow$ constant $|V_{GS-in}| \Rightarrow V_{S-in} \uparrow \Rightarrow |V_{DS-tail}| \downarrow$
 max V_{icm} is when $|V_{DS-tail}| = |V_{dsat-tail}| \Rightarrow V_{icm-max} = 0.76v$

When $V_{icm} \downarrow$, constant $I_D \Rightarrow$ constant $|V_{GS-in}|$ and constant $V_{D-in} = V_{RD} \Rightarrow V_{S-in} \downarrow \Rightarrow |V_{DS-in}| \downarrow$
 min V_{icm} is when $|V_{DS-in}| = |V_{dsat-in}| \Rightarrow V_{icm-min} = -0.43v$

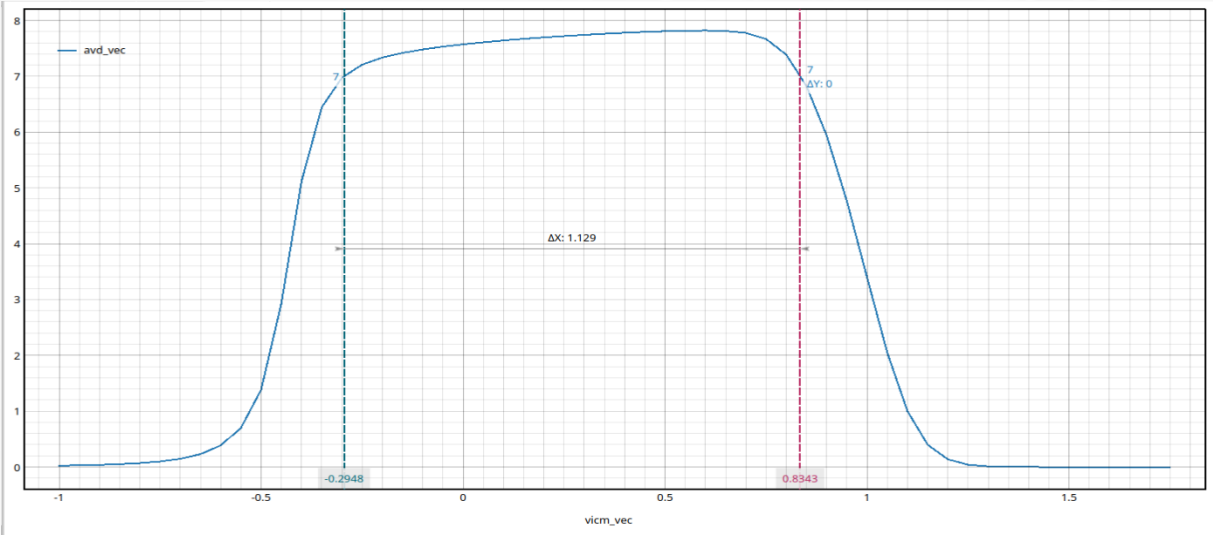
The CMIR analytically we were calculating above $(-0.27 \rightarrow 0.71v) = 0.98$

parameter	CM large signal analysis		
	Simulation	analytical	Error
$V_{icm-max}$	0.76v	0.71v	0.12v
$V_{icm-min}$	-0.43v	-0.27v	0.03v
CMIR	1.19v	0.98v	0.21v

The min V_{icm} is not as expected, why?

When V_{icm} decrease, constant $I_D \Rightarrow$ constant $|V_{GS-in}|$ and constant $V_{D-in} = V_{RD} \Rightarrow V_{S-in}$ decrease
 $\Rightarrow |V_{SB-in}|$ increase $\Rightarrow$ Body effects $\uparrow \Rightarrow |V_{th-in}| \uparrow \Rightarrow (|V_{GS-in}| = V_{S-in} - V_{G-in}) \uparrow$
 that's mean V_S will slowly decrease relative to V_G which reduce the voltage at which the input transistor will go out of saturation $V_{DS} < V_{dsat}$

CM large signal: GBW vs V_{icm}

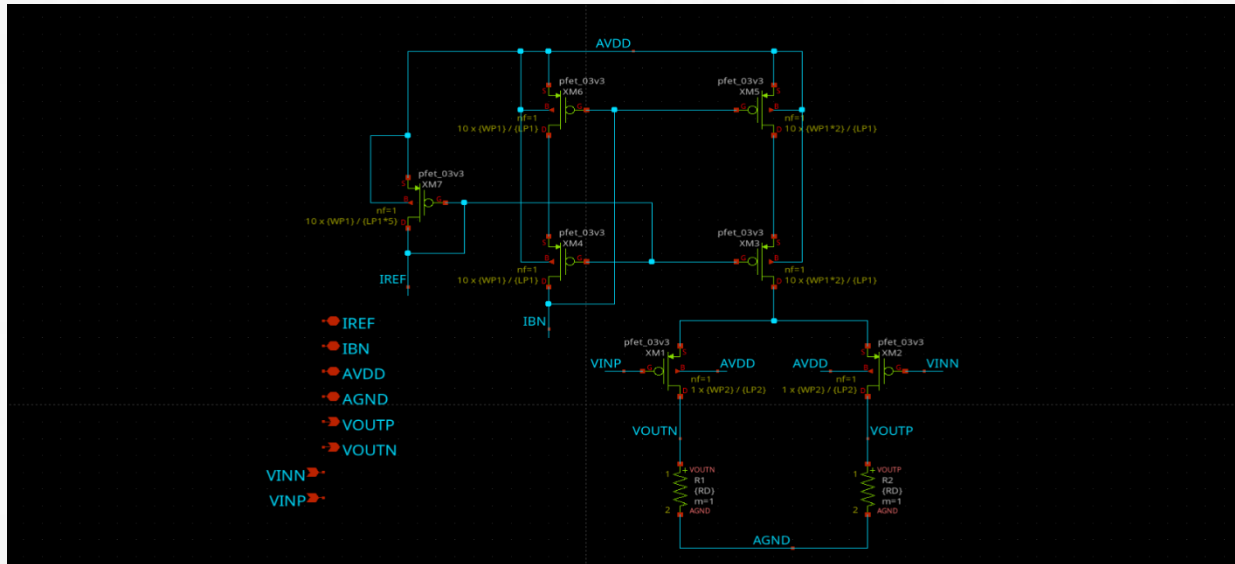


A_{vd} vs V_{icm} in range between 10% drop in the gain ≈ 7

Find the CM input range. Compare with the previous method (V_{dsat}) in a table.

parameter	CM large signal analysis		
	region vs V_{icm}	GBW vs V_{icm}	Error
$V_{icm-max}$	$0.76v$	$0.83v$	$0.07v$
$V_{icm-min}$	$-0.43v$	$-0.3v$	$0.13v$
$CMIR$	$1.19v$	$1.13v$	$0.06v$

Wide Swing Current Mirror



Input pair sizing

Take compliance voltage = $250\text{mV} \Rightarrow V_{CM}^* = 125\text{mV} \Rightarrow \frac{g_m}{I_D} = 16$

Let $V_{DS_{CM}} = V^* + 125\text{mV} = 250\text{mV}$

$\Rightarrow V_{S-in} = V_{DD} - 2V_{DS_{CM}} = 1.8 - 0.5 = 1.3\text{V}$

$V_{sb-in} = 1.8 - 1.3 = 0.5\text{V}$

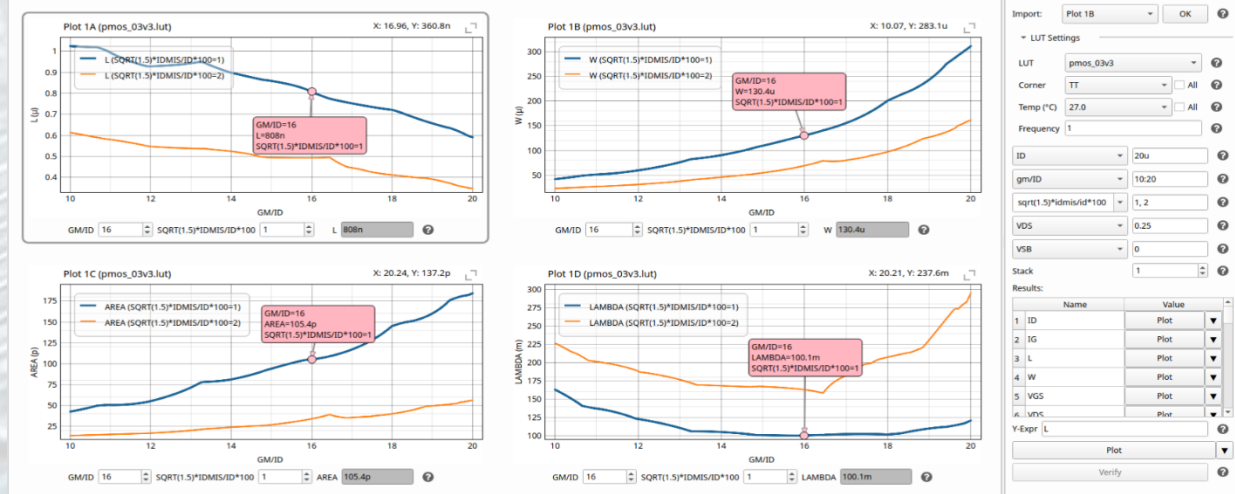
$V_{DS-in} = 1.3 - 20\mu\text{A} \cdot 30\text{K}\Omega = 1.3 - 0.6 = 0.7\text{V}$

$\Rightarrow L = 360\text{nm}, W = 32\mu\text{m}$

From ADT: $V_{GS} = 1\text{V} \Rightarrow CMIL = V_{DD} - 2V_{DS} - V_{GS} = 1.8 - 0.5 - 1 = 0.3\text{V}$

Current mirror sizing

For compliance voltage $\leq 250\text{mV} \rightarrow \frac{g_m}{I_D} \geq 16$, take $\frac{g_m}{I_D} = 16$ to minimize the area. $V_{SB} = 0, V_{DS} = 0.25\text{V}, I_D = 20\mu\text{A}$

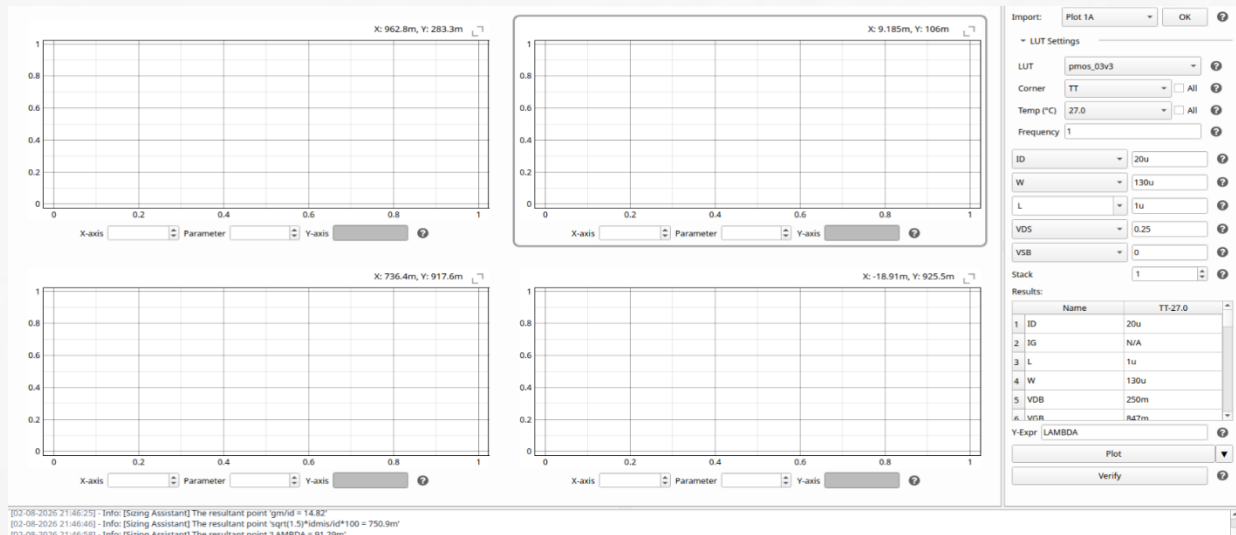


Wide swing current mirror sizing

We will pick **1%** mismatch, since as shown in the λ graph above for 2% mismatch it doesn't have a value for λ that meet the specs that is $\lambda < 0.1$

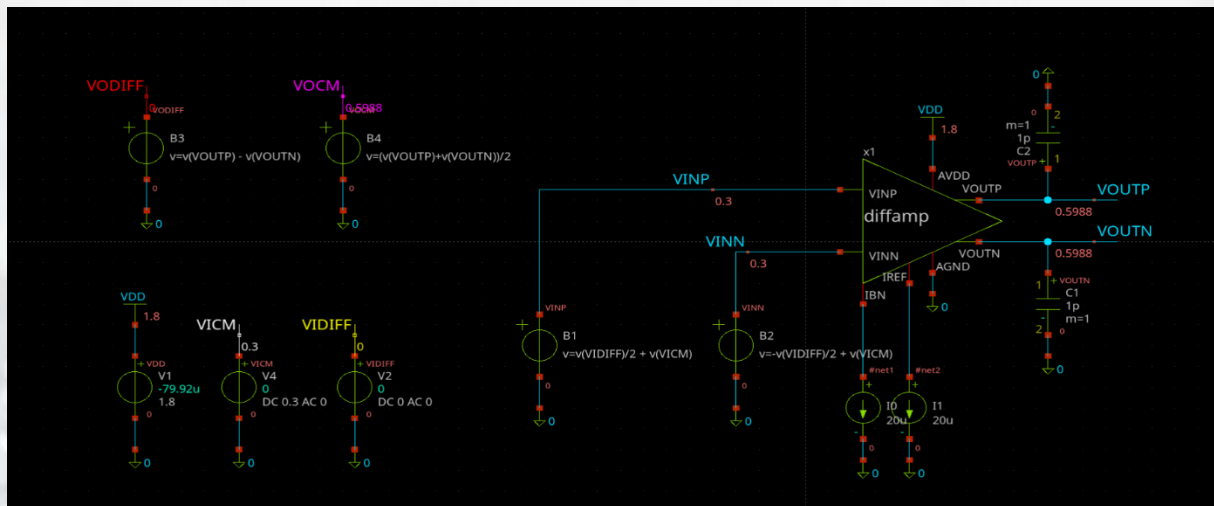
$\Rightarrow$ take $L = 1\mu\text{m}$, $W = 130\mu\text{m}$, for mirror transistor $W = 260\mu$ to get mirroring ratio = 2

We get good compliance voltage in the expense of the Area.

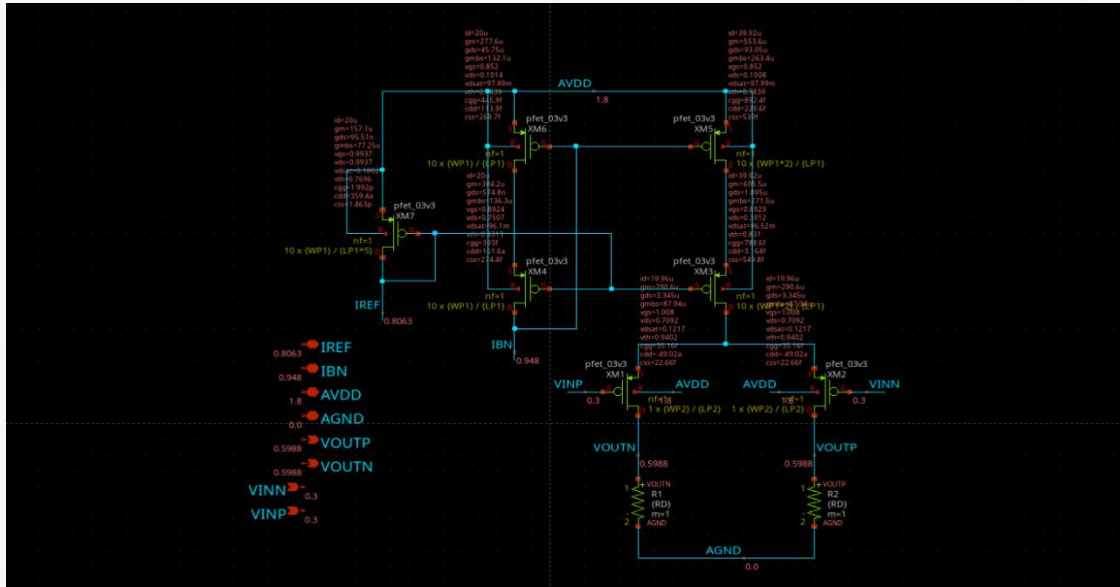


Wide swing current mirror design point

At the selected sizing the mismatch is 0.75% and $\lambda = 0.091 < 0.1$ which is meet our specs.



OP simulation



```

=== MOSFET Operating Points ===
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm7.m0      m.x1.xm6.m0      m.x1.xm5.m0
model       pfet_03v3.14    pfet_03v3.13      pfet_03v3.13
id          2e-05          2e-05             3.99206e-05
vgs         0.993721       0.852035          0.852035
vds         0.99372       0.101352          0.100824
vth         0.769588       0.783942          0.783942
vdsat       0.180193       0.0979932         0.0979932
gm          0.000157101     0.000277636       0.000553562
gds         9.55114e-08     4.57464e-05       9.30498e-05
gmbs        7.72452e-05     0.000132121      0.000263448

BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm4.m0      m.x1.xm3.m0      m.x1.xm2.m0
model       pfet_03v3.13    pfet_03v3.13      pfet_03v3.12
id          2e-05          3.99206e-05       1.99603e-05
vgs         0.892368       0.892896          1.008
vds         0.750683       0.391171          0.709193
vth         0.831265       0.831026          0.940189
vdsat       0.0961024       0.0965178         0.121655
gm          0.000304164     0.000605462       0.000290642
gds         5.7482e-07      1.89459e-06       3.34548e-06
gmbs        0.00013634       0.000271456      8.79363e-05

BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm1.m0
model       pfet_03v3.12
id          1.99603e-05
vgs         1.008
vds         0.709193
vth         0.940189
vdsat       0.121655
gm          0.000290642
gds         3.34548e-06
gmbs        8.79363e-05
  
```

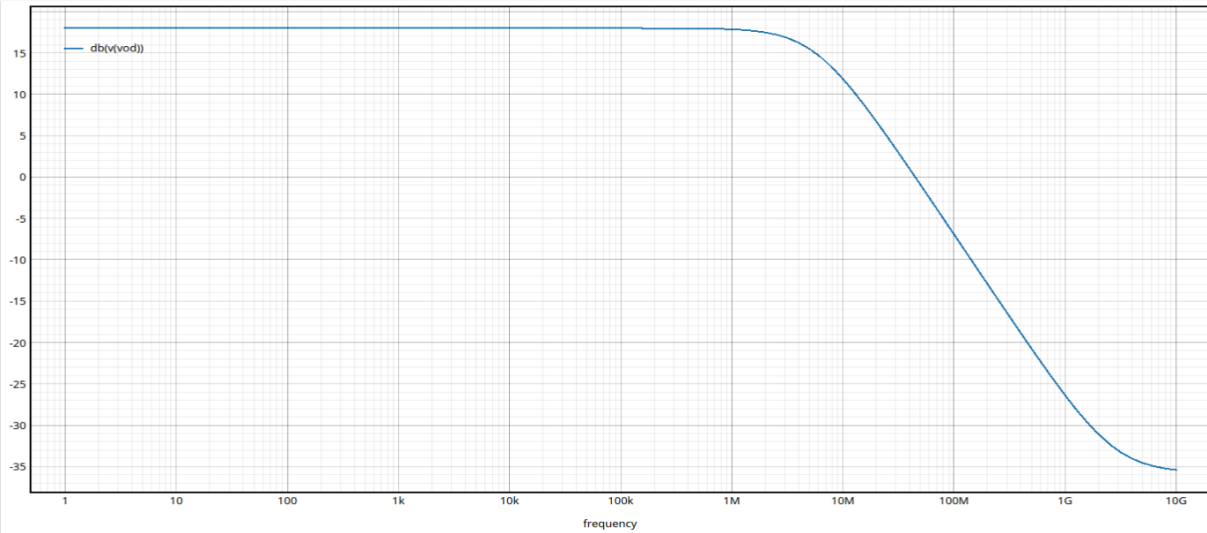
```

M1: gm/id = 14.561 S/A, Vstar = 0.137353 V
M2: gm/id = 14.561 S/A, Vstar = 0.137353 V
M3: gm/id = 15.1667 S/A, Vstar = 0.131868 V
M4: gm/id = 15.2082 S/A, Vstar = 0.131508 V
M5: gm/id = 13.8666 S/A, Vstar = 0.144232 V
M6: gm/id = 13.8818 S/A, Vstar = 0.144073 V
M7: gm/id = 7.85506 S/A, Vstar = 0.254613 V
  
```

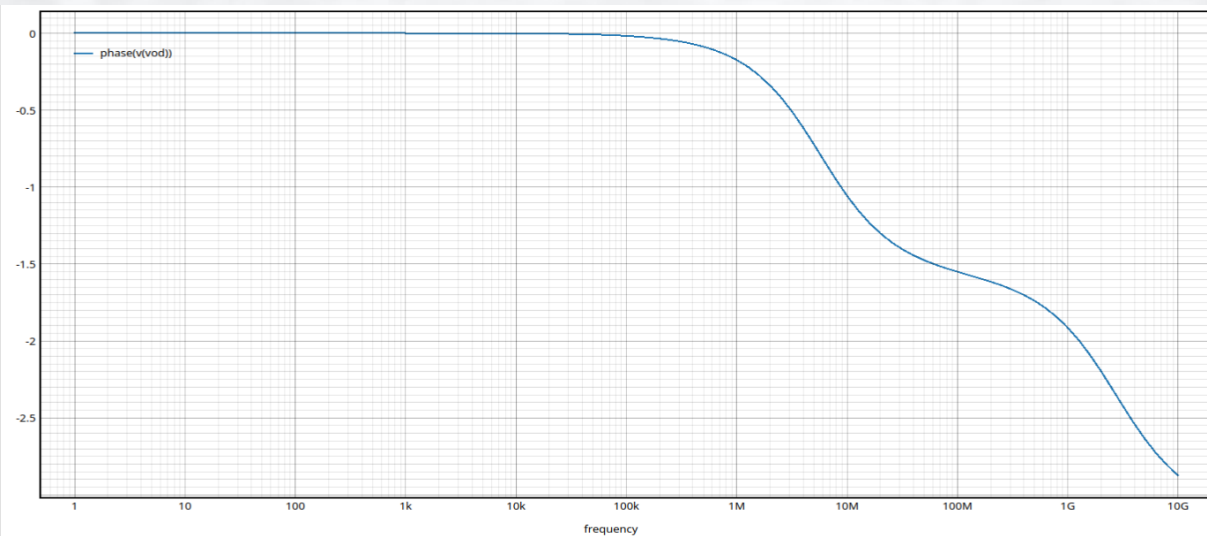
AC analysis

```
av_diff_db      = 1.797875e+01
av_diff_mag     = 7.923874e+00
bw_diff        = 5.707869e+06
av_cm_db       = -5.367975e+01
av_cm_mag      = 2.070202e-03
meas_ac_bw_cm  when vmag(vocm)=1.463854e-03 fall=1 failed!

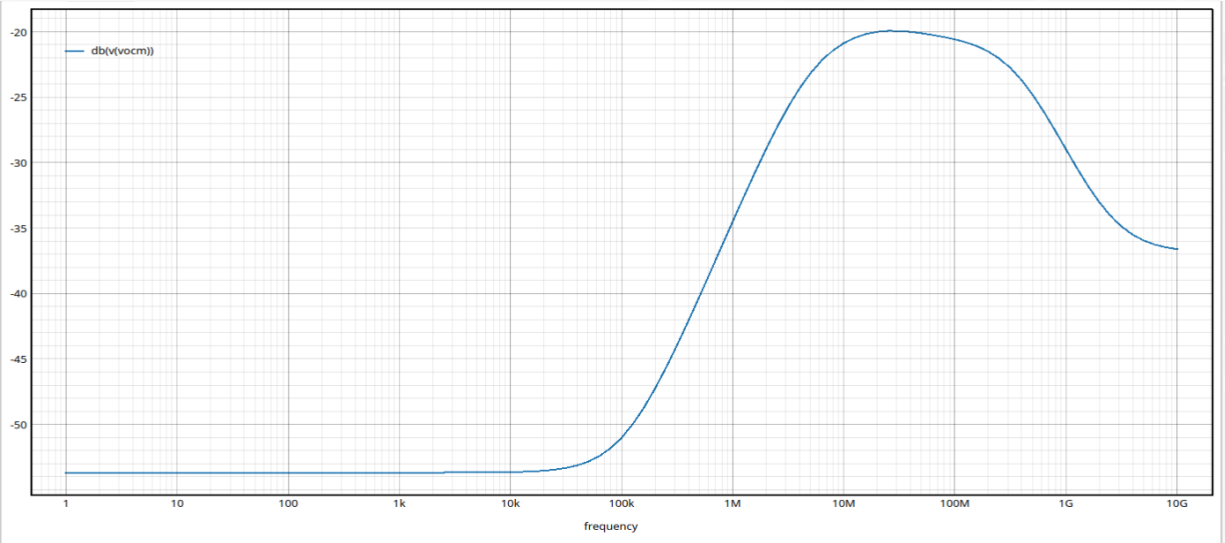
=== AC Results ===
Av Diff: 17.9787 dB, BW: 5.70787E+06 Hz
Av CM: -53.6798 dB, BW: Hz
CMRR: 71.6585 dB
```



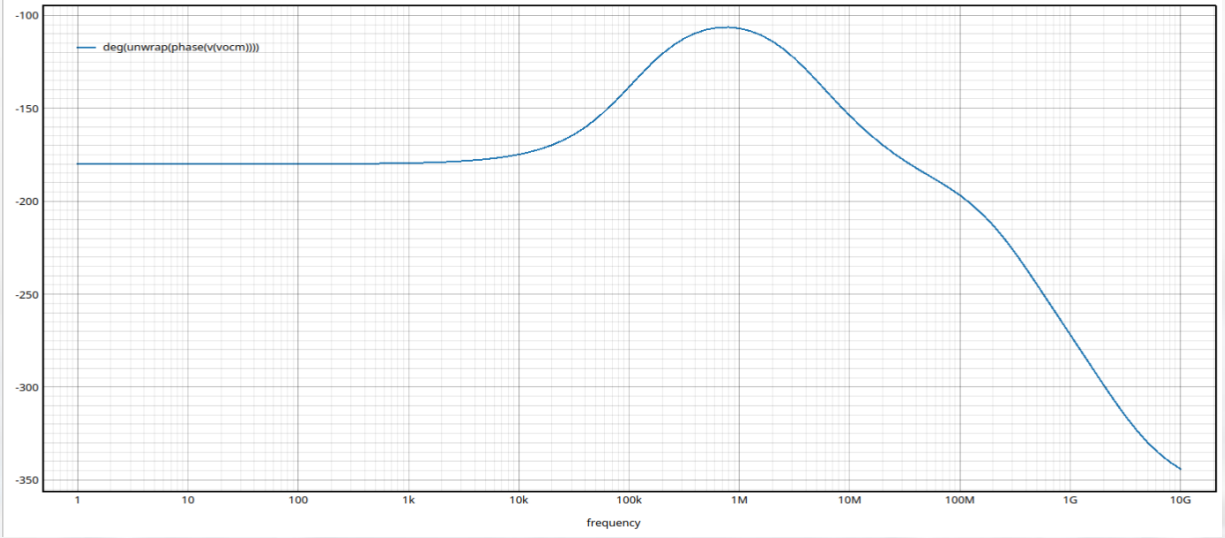
Bode plot: A_{vd} mag



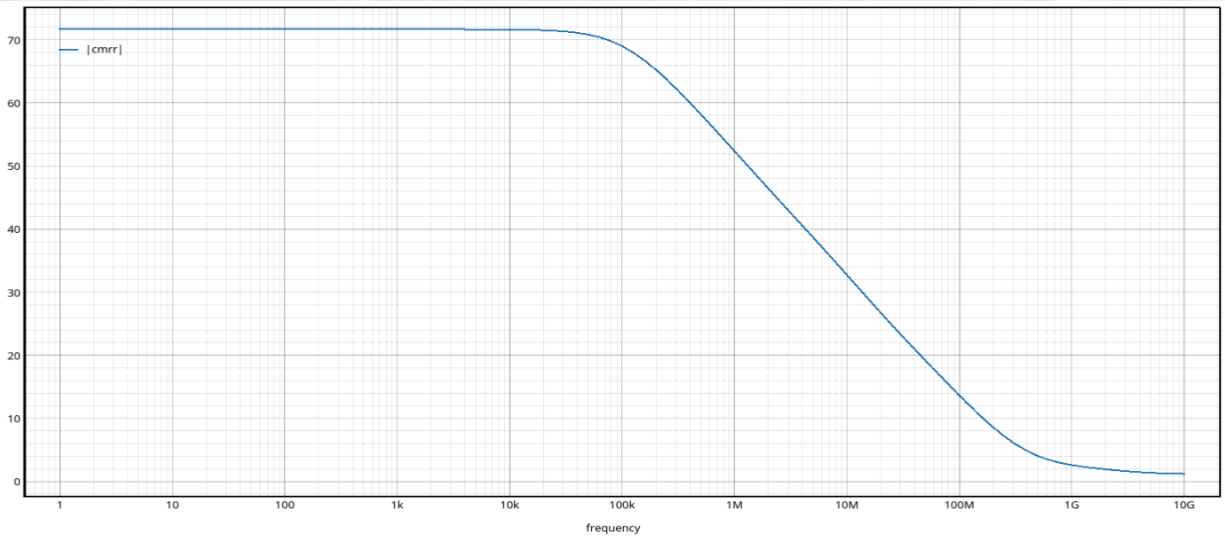
Bode plot: A_{vd} phase



Bode plot: A_{vcm} mag

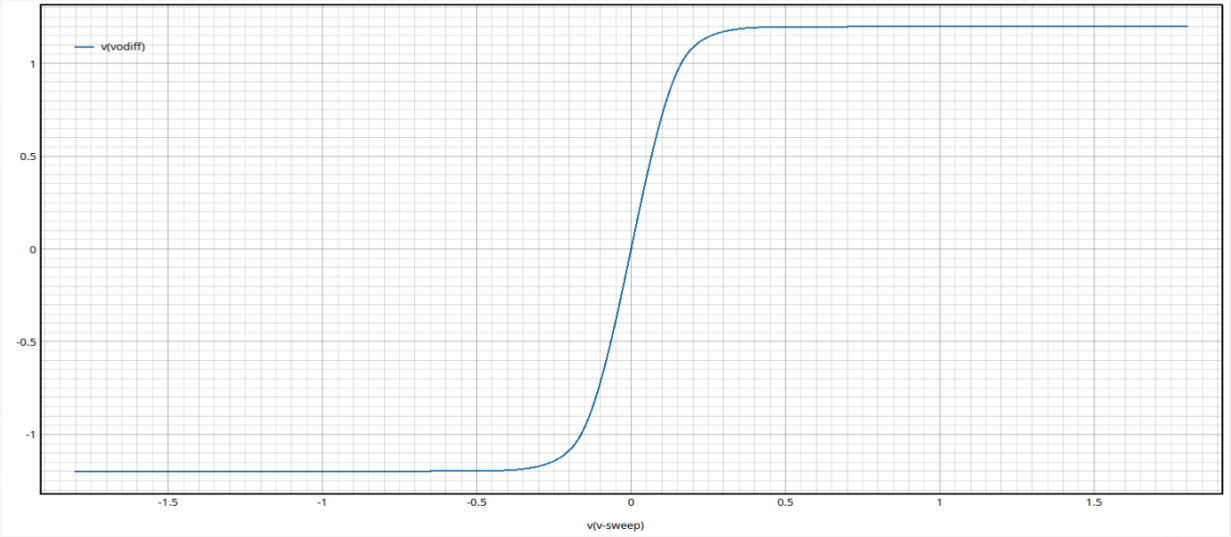


Bode plot: A_{vcm} phase

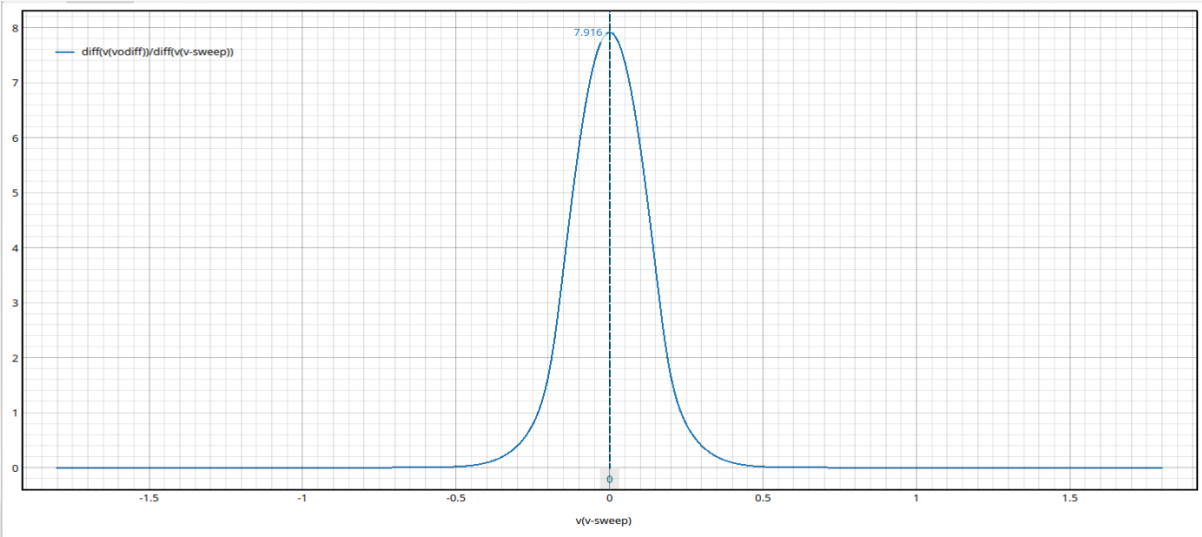


CMRR

DC analysis



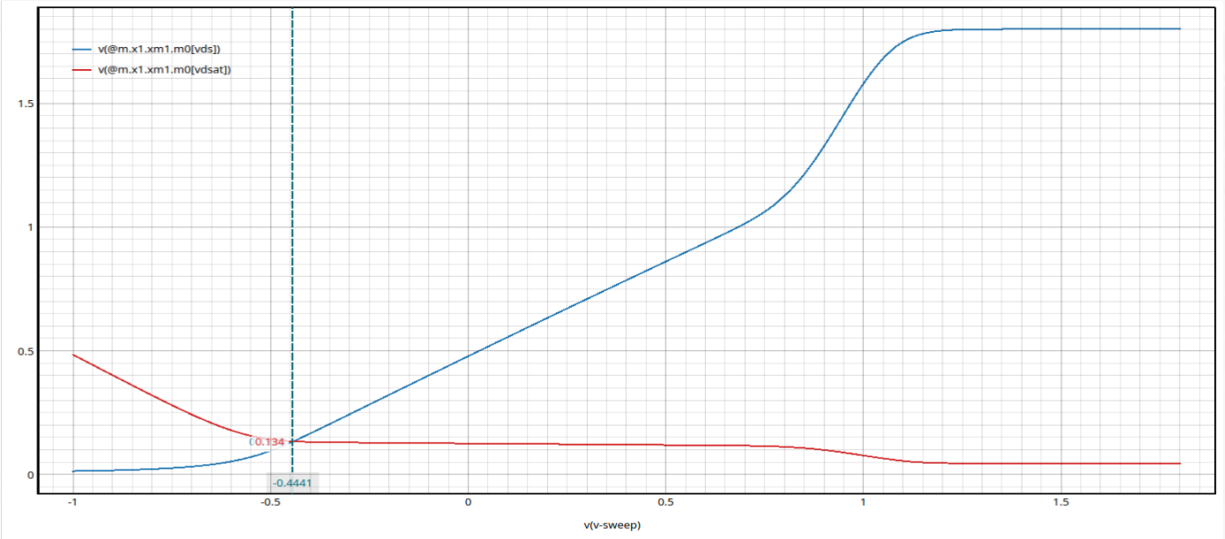
V_{od} vs V_{id}



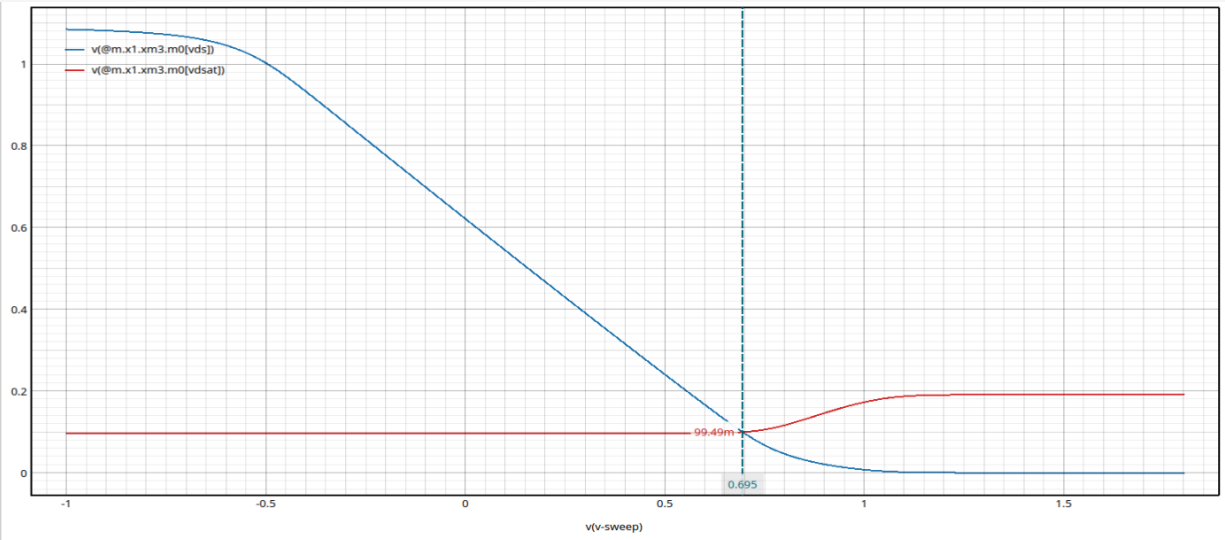
V_{od} derivative vs V_{id}

Diff Large Signal Analysis Results:
Max Output: 1.19808 V
Min Output: -1.19808 V

parameter	Differential large signal analysis		
	Simple CM	Wide swing CM	Error
Max output	1.188v	1.198v	0.01
Min output	-1.188v	-1.198v	0.01

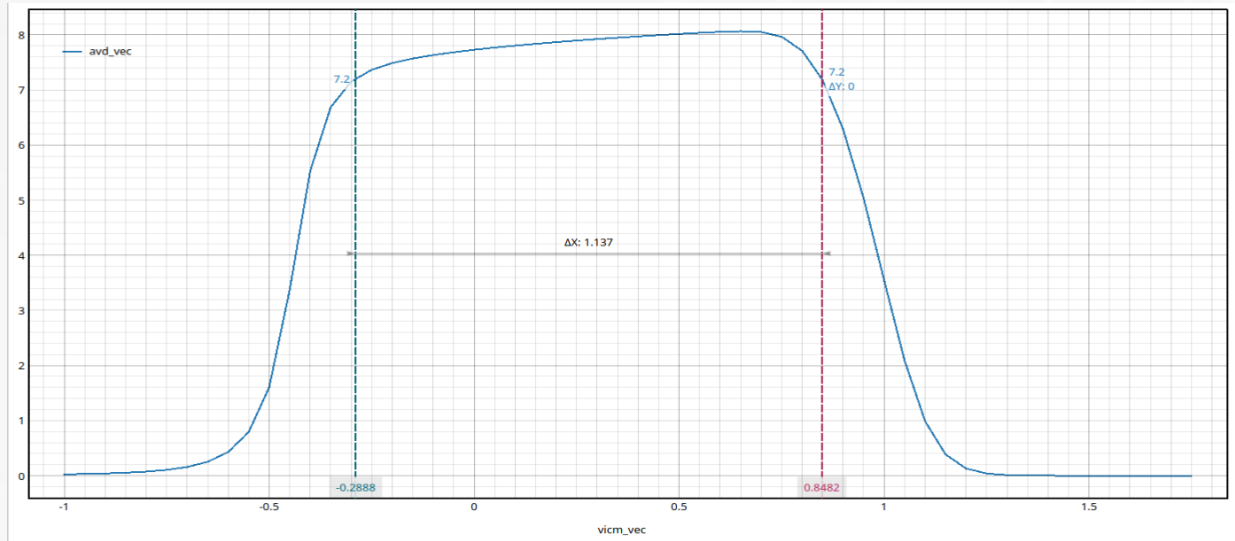


■ V_{DS} and ■ V_{dsat} vs V_{icm} for the input pair



■ V_{DS} and ■ V_{dsat} vs V_{icm} for the tail current source

parameter	CM large signal analysis		
	Simple CM	Wide swing CM	Error
$V_{icm-max}$	$0.76v$	$0.7v$	$0.06v$
$V_{icm-min}$	$-0.43v$	$-0.44v$	$0.01v$
$CMIR$	$1.19v$	$1.14v$	$0.05v$



parameter	CM large signal analysis		
	Simple CM	Wide swing CM	Error
$V_{icm-max}$	$0.83v$	$0.85v$	$0.02v$
$V_{icm-min}$	$-0.3v$	$-0.29v$	$0.01v$
$CMIR$	$1.13v$	$1.14v$	$0.01v$

Comparing the above results with the simple current mirror case we find that:

Both almost have the same differential gain

In case using wide swing current mirror:

it has better (smaller) common mode gain only at low frequency

it has better (larger) CMRR only at low frequency

more accurate results to the analytical